1 Hexagon Inner Area

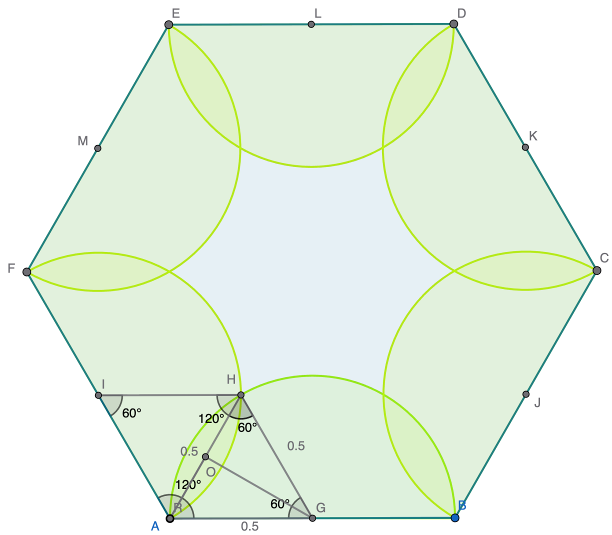

Figure 1: Diagram

Area of blue inner region
= Area of Hexagon - 6 * Area of Semicircle + 6 * Area of overlap regions

$$A_{\text{Hexagon}} = \frac{3\sqrt{3}}{2}.1^2 = \frac{3\sqrt{3}}{2}$$

$$A_{\text{Semicircle}} = \pi.(\frac{1}{2})^2.\frac{1}{2} = \frac{\pi}{8}$$

$\angle FAB = 120°$
AH bisects $\angle FAB$ by symmetry
Therefore $\angle HAG = 60°$
AG = AH, so $\angle AHG = 60°$
Therefore $\triangle AHG$ is equilateral
$A_{\triangle AHG} = \frac{\sqrt{3}}{4}.(\frac{1}{2})^2 = \frac{\sqrt{3}}{16}$
$A_{\text{arc } AHG} = \pi.(\frac{1}{2})^2.\frac{1}{6} = \frac{\pi}{24}$
$A_{\text{circle intersection}} = 2(A_{\text{arc } AHG} - A_{\triangle AHG})$
$= 2(\frac{\pi}{24} - \frac{\sqrt{3}}{16})$
$= \frac{\pi}{12} - \frac{\sqrt{3}}{8}$

$$
\begin{aligned}
A_{\text{shaded}} &= \tfrac{3\sqrt{3}}{2} - \tfrac{6\pi}{8} + 6\left(\tfrac{\pi}{12} - \tfrac{\sqrt{3}}{8}\right) \\
&= \tfrac{3\sqrt{3}}{2} - \tfrac{3\pi}{4} + \tfrac{6\pi}{12} - \tfrac{6\sqrt{3}}{8} \\
&= \tfrac{3\sqrt{3}}{2} - \tfrac{3\pi}{4} + \tfrac{\pi}{2} - \tfrac{3\sqrt{3}}{4} \\
&= \tfrac{6\sqrt{3}}{4} - \tfrac{3\sqrt{3}}{4} - \tfrac{3\pi}{4} + \tfrac{2\pi}{4} \\
&= \tfrac{3\sqrt{3}}{4} - \tfrac{\pi}{4} \\
&= \boxed{\frac{3\sqrt{3} - \pi}{4}}
\end{aligned}
$$